

Week 2 Deliverable

Value Analysis

Project / Product

Project: AlgaNova
Group: 9
Revision: 1
Date: 7/24/2026

Transferred data from Requirements List - No editing here!						Concept 1			Concept 2	
ID	Requirement	Target / Criterion	Unit	Demand / Wish	Weight	Degree of Fulfillment	value [0...10]	weighted value	Degree of Fulfillment2	value [0...10]2
1	Water depth operation	0 - 1	m	D	0.07	0-0.2m	2	0.14	0-0.5m	5
2	Operating speed	~1	Km/h	D	0.05	1->6 Km/h	9	0.45	1->4 Km/h	8
3	Noise emission	~60	dB	W	0.03	~5dB	10	0.3	60 -> 70dB	5
4	Energy consumption	<500	W	D	0.06	0W	9	0.54	250W	8
5	Operating Duration	4->6	h	D	0.04	3h	7	0.28	4h	8
6	Algae storage capacity	200 -> 250	Kg	D	0.05	80Kg	2	0.1	100Kg	7
7	Machine weight	<500	Kg	W	0.07	Canoe + Gear + 2Crew =210Kg	10	0.7	Machine + Crew ~450Kg	7
8	Regular Maintenance (Cleaning)	100	Euros	D	0.04	10 Euros	10	0.4	100 Euros	10
9	Harmful emissions	None	None	D	0.05	Zero Emission	10	0.5	None	10
10	Corrosion Resistance	Visual Inspection	None	W	0.03	Material mostly made of plastic	9	0.27	On a scale from 1-10: we put it at 7	7
11	Compact design (Surface Area)	<5	m^2	W	0.04	~3.25	8	0.32	~4	10
12	Attractive appearance	public opinion on design	None	W	0.03	On a scale from 1-10: we put it at a 7	7	0.21	On a scale from 1-10: we put it at a 8	8
13	Durability	10	years/w arranty	D	0.03	Check	10	0.3	Check	10
14	Machine Safety & Electrical and Radio Regulations	Class 1	None	D	0.03	Check	10	0.3	Check	10
15	Battery Safety Regulations	UN 38.3 Certification.	None	D	0.03	check	10	0.3	Check	10
16	Comply to the Federal Waterway Permit (Strom- und Schifffahrtpolizeiliche Genehmigung)	Conform to legal requirements	None	D	0.04	check	10	0.4	Check	10
17	Water Use Permit (Wasserrechtliche erlaubnis)	Conform to legal requirements	None	D	0.04	check	10	0.4	Check	10
18	Biodiversity & Habitat Protection (Naturschutzrecht)	Conform to legal requirements	None	D	0.03	Check	10	0.3	Check	10
19	Scalability	Adaptable to Canal width	m	W	0.02	Check	10	0.2	Check	10
20	Water proof	IPX6	None	D	0.04	Check	10	0.4	Check	10
21	Must not block canal Navigation	Stoppage time of other simultaneously occurring activities	min	D	0.03	Check	10	0.3	Check	10
22	Preserve canal infrastructure	No reconstruction nor destruction of the currents canal's infrastructure	None	D	0.05	Check	10	0.5	Check	10

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Value Scale: 0 (absolut useless solution) - 10 (ideal solution)

	Concept 3		
weighted value2	Degree of Fulfillment3	value [0...10]3	weighted value3
0.35	0-0.5m	5	0.35
0.4	1->4 Km/h	8	0.4
0.15	60 -> 80dB	4	0.12
0.48	450W	6	0.36
0.32	4h	8	0.32
0.35	100Kg	5	0.25
0.49	Machine ~450Kg	7	0.49
0.4	150 Euros	8	0.32
0.5	None	10	0.5
0.21	On a scale from 1-10: we put it at 7	6	0.18
0.4	4 - >5	10	0.4
0.24	On a scale from 1-10: we put it at a 8	8	0.24
0.3	Check	10	0.3
0.3	Check	10	0.3
0.3	Check	10	0.3
0.4	Check	10	0.4
0.4	Check	10	0.4
0.3	Check	10	0.3
0.2	Check	10	0.2
0.4	Check	10	0.4
0.3	Check	10	0.3
0.5	Check	10	0.5

23	Water Resistant	Should Operable in hot,sunny,moist and rainy weather conditions	None	D	0.02	Check	10	0.2	Check	10
24						(20->25)Kg	3		(~50)Kg	5
25				Total Weight	0.92			0		
26				Check	Weights must sum to 1.000		Σ	7.81		Σ

0.2	Check	10	0.2
	(~70)Kg	6	
0			0
7.89		Σ	7.53